SOURCE CODE: -

```
#include <stdio.h>

#include <string.h>

// Binary Search function

int binarySearch(char files[][50], int n, char target[])

{

    int low = 0;

    int high = n - 1;


    while (low <= high)

    {

        // Find the middle position

        int mid = low + (high - low) / 2;

        int comparison = strcmp(files[mid], target);

        // File found

        if (comparison == 0)

        {

            return mid;

        }

        // Target comes after the middle element

        if (comparison < 0)

        {

            low = mid + 1;

        }


        // Target comes before the middle element
```

```c
        else
        {
            high = mid - 1;
        }
    }
    return -1;
}
int main()
{
    char files[][50] = {
        "Algorithms.pdf",
        "Database.docx",
        "Networks.pdf",
        "OperatingSystem.pdf",
        "Programming.c",
        "ResearchPaper.pdf",
        "SystemDesign.docx",
        "WebDevelopment.html"
    };
    int n = sizeof(files) / sizeof(files[0]);
    char target[50];

    printf("File System Search using Binary Search\n");
    printf("========================================\n");
    printf("\nAvailable Files:\n");
    for (int i = 0; i < n; i++)
    {
```

```
        printf("%d. %s\n", i + 1, files[i]);
    }
    printf("\nEnter file name to search: ");
    scanf("%49s", target);
    int result = binarySearch(files, n, target);
    if (result != -1)
    {
        printf("\nFile found!\n");
        printf("File: %s\n", files[result]);
        printf("Index: %d\n", result);
    }
    else
    {
        printf("\nFile not found.\n");
    }
    return 0;
}
```

Sample Output

```
File System Search using Binary Search
=======================================

Available Files:
1. Algorithms.pdf
2. Database.docx
3. Networks.pdf
4. OperatingSystem.pdf
5. Programming.c
6. ResearchPaper.pdf
7. SystemDesign.docx
8. WebDevelopment.html

Enter file name to search: ResearchPaper.pdf

File found!
File: ResearchPaper.pdf
Index: 5
```